One student can be selected in 10 ways among 10 boys.

One student can be selected in 8 ways among 8 girls

Hence the jobs can be performed in 10 + 8 = 18 ways

This problem can be solved by applying Addition principle of counting.

Definition 2.2

If there are two jobs, each of which can be performed independently in m and n ways respectively, then either of the two jobs can be performed in (m+n) ways. This is called addition principle of counting.

Example 2.14

There are 6 books on commerce and 5 books on accountancy in a book shop. In how many ways can a student purchase either a book on commerce or a book on accountancy?

Solution

Out of 6 commerce books, a book can be purchased in 6 ways.

Out of 5 accountancy books, a book can be purchased in 5 ways.

Then by addition principle of counting the total number of ways of purchasing any one book is 5+6=11 ways.

Exercise 2.2

1. Find x if $\frac{1}{6!}+\frac{1}{7!}=\frac{x}{8!}$.
2. Evaluate $\frac{n!}{r!(n-r)!}$ when $n = 5$ and $r = 2$.
3. If $(n+2)! = 60[(n-1)!]$, find n
4. How many five digits telephone numbers can be constructed using the digits 0 to 9 if each number starts with 67 with no digit appears more than once?
5. How many numbers lesser than 1000 can be formed using the digits 5, 6, 7, 8 and 9 if no digit is repeated?

2.2.4 Permutation

Suppose we have a fruit salad with combination of "APPLES, GRAPES & BANANAS". We don't care what order the fruits are in. They could also be "bananas, grapes and apples" or "grapes, apples and bananas". Here the order of mixing is not important. Any how, we will have the same fruit salad.

But, consider a number lock with the number code 395. If we make more than three attempts wrongly, then it locked permanently. In that situation we do care about the order 395. The lock will not work if we input 3-5-9 or 9-5-3. The number lock will work if it is exactly 3-9-5. We have so many such situations in our practical life. So we should know the order of arrangement, called Permutation.

Definition 2.3

The number of arrangements that can be made out of n things taking r at a time is called the number of permutation of n things taking r at a time.

For example, the number of three digit numbers formed using the digits 1, 2, 3 taking all at a time is 6.

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